

Outcome of Ruthenium-106 Brachytherapy for Uveal Melanoma and Retinal Vascular Tumors: A Retrospective Study

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Abstract

Purpose: To evaluate the efficacy and safety of brachytherapy in patients with uveal melanoma and retinal vascular tumors.

Methods: Patients who received Ruthenium-106 (Ru-106) plaque brachytherapy at the Torfeh Medical Center between March 2016 and March 2022 were examined in this retrospective case-series. We collected clinical data on tumor features, treatment details, and patient outcomes. These data were analyzed using SPSS software. Tumor regression was the primary outcome that was monitored, with complications being evaluated as secondary outcomes.

Results: Of the 80 patients, 55 had melanoma, 21 had vasoproliferative tumor (VPT), and 4 had retinal capillary hemangioma (RCH). Tumor regression was observed in 74.5% of melanoma cases and 100% of VPT/RCH cases. Significant relationships were found between tumor characteristics, such as thickness and dose rate at the apex, and treatment outcomes. Complications included radiation retinopathy and cataract formation, with recurrence rates of 16.4% in melanoma and 7.7% in VPT/RCH.

Conclusions: Ru-106 plaque brachytherapy is effective in treating various intraocular tumors, particularly VPT/RCH. Tumor thickness significantly influences treatment outcomes, emphasizing the need for tailored approaches in brachytherapy planning. However, small sample sizes for specific tumor types limited the generalization of the outcomes. Future research should focus on refining techniques and mitigating complications to improve patient outcomes in ocular oncology.

Keywords: Brachytherapy, Intraocular tumor, Outcome

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Submitted: 18-Nov-2024; **Revised:** 06-Oct-2025; **Accepted:** 17-Oct-2025; **Published:** 19-Dec-2025

INTRODUCTION

Ocular tumors, comprising both primary and metastatic cancers, represent a significant health burden due to their potential to threaten vision and life.^{1,2} According to a global analysis of cancer registries, the age-standardized rates of eye cancer incidence and mortality in 2020 were 0.49 and 0.08/100,000 globally, respectively.¹ A study utilizing data from the Iranian National Population-based Cancer Registry revealed that the overall incidence proportion of ocular cancers

in Iran from 2004 to 2016 was 16.04/100,000 population. Specifically, this registry documented a 6% prevalence of intraocular tumors among patients with cancer.³

Plaque brachytherapy has long been recognized as a pivotal treatment modality for medium-sized uveal melanoma, marking its significance in the realm of ocular oncology.^{4,5} This technique harnesses the power of localized radiation,

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How to cite this article: Karimi S, Ashrafi S, Hosseini Imeni SM, Siavashpour Z. Outcome of Ruthenium-106 brachytherapy for uveal melanoma and retinal vascular tumors: A retrospective study. J Curr Ophthalmol 2025;37:247-55.

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DOI:
10.4103/joco.joco_248_24

meticulously administered to the eye through a radioactive plaque that is surgically attached directly over the intraocular tumor. Such a targeted approach ensures the delivery of therapeutic radiation doses directly to the tumor site while sparing the adjacent healthy tissues. This method not only serves as a cornerstone in the preservation of eye structure but also plays a crucial role in maintaining vision, offering a valuable alternative to the more radical option of enucleation for patients with intraocular tumors.^{6,7}

Introduced in the early 20th century by Moore,⁸ plaque brachytherapy was initially developed to address intraocular tumors, predominantly retinoblastoma and choroidal melanoma. Over the years, the evolution of this technique has seen the utilization of various radioisotopes. The preference for Ruthenium-106 (Ru-106), a beta emitter, over gamma emitters is attributed to its ability to produce a steeper radiation gradient. This characteristic significantly minimizes the radiation exposure to the surrounding healthy ocular structures while maximizing the dose to the tumor base, underlining the nuanced advancements in plaque brachytherapy that have enhanced its efficacy and safety as a treatment option.^{9,10}

Previous studies exploring the efficacy and safety of Ru-106 plaque brachytherapy for intraocular tumors have yielded a broad spectrum of results, highlighting both the potential and limitations of this treatment modality. While many reports underscore the high rates of local tumor control and vision preservation afforded by Ru-106, discrepancies arise in the context of long-term safety and complication rates. Some studies indicate a relatively low incidence of adverse effects, such as radiation retinopathy and neovascular glaucoma, suggesting that Ru-106 is a safe option for patients.¹¹⁻¹³ Conversely, others document a higher occurrence of these complications, leading to concerns about the long-term ocular health and quality of life for treated individuals.⁶

Intraocular tumors exhibit significant biological diversity, including various benign and malignant forms that demonstrate distinct growth patterns, metastatic capabilities, radiosensitivity, and prognostic outcomes. Uveal melanomas exhibit aggressive local behavior and a significant metastatic risk,¹⁴ whereas choroidal hemangiomas are benign vascular lesions characterized by distinct treatment responses.¹⁵ The observed differences indicate that outcomes and complications associated with Ru-106 brachytherapy may differ markedly across tumor types. Previous studies infrequently consider this biological heterogeneity, which restricts the generalizability of their findings across various intraocular tumors.

This study included a diverse group of intraocular tumors to evaluate the efficacy and safety of Ru-106 plaque brachytherapy across various biological entities. This study seeks to evaluate the potential of Ru-106 as an effective and versatile treatment option across various tumor types and biological behaviors. Moreover, most of these investigations have focused predominantly on a single type of tumor, often choroidal melanoma, limiting the applicability of their findings

to the diverse spectrum of intraocular tumors. While Ru-106 plaque brachytherapy remains a valuable tool in the arsenal against intraocular tumors, the noted discrepancies. The present study was conducted with the aim of evaluating the efficacy and safety of brachytherapy in patients with uveal melanoma and retinal vascular tumors.

METHODS

In this retrospective consecutive case series, we evaluated the outcomes of Ru-106 plaque brachytherapy in 80 patients manifesting symptoms with ocular tumors. These patients were treated at the Torfeh Medical Center from March 2016 to March 2022. To be included in the study, all patients were required to have a follow-up period of 6 months or longer. In addition, the tumor diameter was required to be <10 mm. For cases where the tumor exceeded this size and the patient did not consent to enucleation, brachytherapy was administered, and these patients were subsequently included in the study. In addition, patients who previously underwent photodynamic therapy or transpupillary thermotherapy (TTT) but experienced treatment failure were included. However, tumors located in the peripapillary region or posterior pole were excluded due to the increased risk of metastasis and the technical limitations associated with plaque placement in these areas; such patients underwent enucleation based on institutional protocol. The research protocol was approved by the Research Ethics Committees of the Vice Chancellor in Research Affairs affiliated to the Shahid Beheshti University of Medical Sciences, Tehran, Iran (IR.SBMU.RETECH.REC.1400.202). Prior to brachytherapy, written informed consent was secured from each participant, confirming their understanding and agreement to the treatment procedure.

All participants underwent comprehensive ophthalmic assessments, which included slit-lamp biomicroscopy, indirect ophthalmoscopy, and advanced ophthalmic imaging techniques such as fundus photography (Topcon, Tokyo, Japan), ultrasonography (UltraScan Imaging System, Alcon Laboratories), and Spectralis OCT (Heidelberg Engineering, Heidelberg, Germany). Follow-up evaluations were systematically conducted every 4 months in the first 2 years postplaque radiotherapy, with subsequent assessments every 6 months. This study collected data encompassing demographic information (age, sex, and the eye under study) and baseline clinical parameters, including best-corrected visual acuity (BCVA), intraocular pressure (IOP), and detailed tumor characteristics (thickness in millimeters, type, and location). Follow-up visits entailed measurements of BCVA and IOP, in addition to monitoring for any adverse effects attributable to the radiotherapy, by a single experienced retina specialist. Furthermore, specific details related to the plaque radiotherapy, such as the duration of treatment (in hours), plaque configuration (CCA, CCB, CGD, COB, CIA models provided by Eckert and Ziegler BEBIG, Berlin, Germany), decay factor, and the total radiation dose (in Gray [Gy])

alongside the dose rate (Gray per hour) delivered to the tumor apex and sclera, were documented.

For the plaque brachytherapy procedure, we utilized the Ru-106 ophthalmic applicator (BEBIG, Berlin, Germany), tailoring the size and shape of the plaque to match the dimensions and location of the tumor. The operative site was accurately marked in the preoperative setting. Patients were then taken to the operating room, where the chosen type of anesthesia, general anesthesia, was administered. The surgery commenced with the sterile draping of the patient's correct eye and the placement of a lid speculum to maintain eyelid separation. Indirect ophthalmoscopy was performed, followed by conjunctival peritomy at the needed aspects of the eye. If necessary, the rectus muscles were secured using a muscle hook and silk 5-0 sutures for precise localization, facilitated by transillumination and indirect ophthalmoscopy. In case the tumor was located under the belly or the insertion of a muscle, the disinsertion and hang-back procedure was used. A re-sterilizable dummy plaque was then positioned and temporarily sutured with Mersilene 5-0. The actual Ru-106 was affixed with mersilene 5-0 sutures, followed by closure of the conjunctiva with Vicryl 7-0. Subconjunctival injections of betamethasone 4 mg and cefazolin 100 mg were administered before the eye was patched. The targeted radiation dose for each patient was determined by a single radiotherapist (ZS), and the plaque was removed postdelivery of the prescribed dose. Both the insertion and removal procedures were conducted by a retina specialist (SK) under general anesthesia for all patients.

The primary clinical outcomes assessed in this study were categorized as tumor regression and the requirement for re-treatment. Regression was defined as a decrease or no change in tumor size, based on the criteria described by Shields *et al.*¹⁶ Beyond these outcomes, comprehensive examinations were conducted at each patient visit to monitor for potential complications. These included the presence of subretinal fluid, cystoid macular edema detectable by optical coherence tomography, and clinically observable conditions such as radiation retinopathy, maculopathy, papillopathy, neovascular glaucoma, cataract formation, and scleral necrosis. In cases where retreatment was deemed necessary (increase in thickness, dimension, SRF formation, new seeding), the chosen modality, ranging from chemotherapy, thermotherapy, to interventions such as argon or diode laser therapy, intravitreal bevacizumab, or cryotherapy, was tailored to the patient's specific needs and the clinical scenario at hand.

Statistical analysis

For the analysis of data in this study, SPSS software version 16 (SPSS Inc., IBM, Chicago, IL, USA) served as the analytical tool. Quantitative variables were summarized using mean and standard deviation, while frequencies (%) described qualitative variables. The comparison of qualitative variables across different groups was performed using the Chi-square test. Conversely, the Mann-Whitney U test was employed for the comparative analysis of quantitative variables.

Receiver operator characteristic (ROC) curve analysis was employed to determine the cutoff values for categorizing tumor length and tumor diameter variables. ROC analysis was chosen because it enables the identification of threshold values that best discriminate between different clinical outcomes (regression vs. progression) based on sensitivity and specificity. Based on this analysis, the cutoff for tumor thickness was established at 6.40 mm, with a sensitivity of 0.67 and a specificity of 0.78. Similarly, the cutoff for tumor length was determined to be 9.75 mm, with a sensitivity of 0.80 and a specificity of 0.56.

RESULTS

In this study, a total of 80 patients were included, comprising 55 with melanoma, 21 with vasoproliferative tumor (VPT), and 4 with retinal capillary hemangioma (RCH) [Figure 1a]. The melanoma group and the VPT/RCH group (categorized based on vascular nature) were examined separately. The average age of patients in the melanoma group was 57.87 ± 16.19 years, and in the VPT/RCH group, it was 35.84 ± 10.63 years [Table 1].

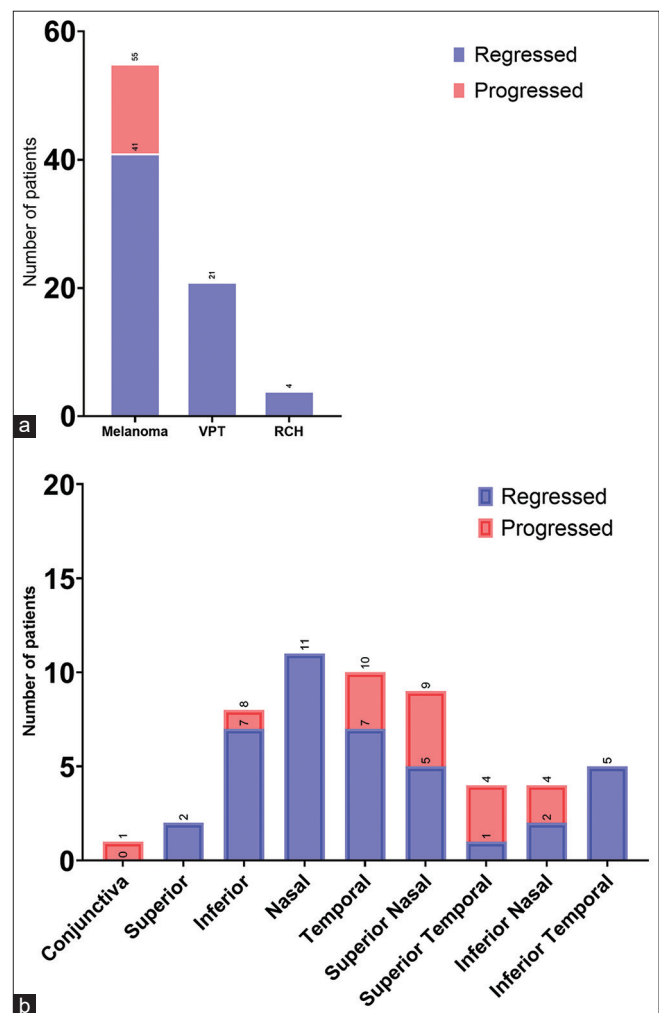


Figure 1: Tumor prevalence and distribution of outcomes by tumor type (a) and location in the melanoma group (b)

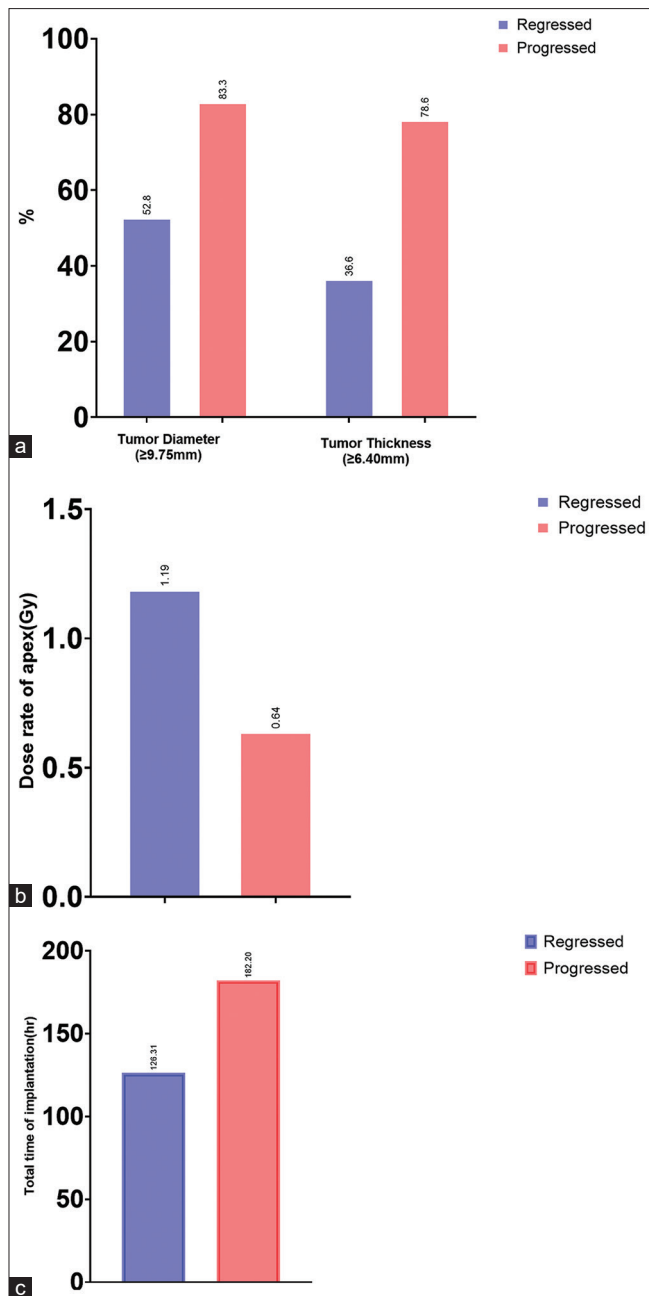


Figure 2: Mean tumor thickness and diameter (a), mean dose rate at the apex (b), mean total time of implantation (c), in the melanoma group, separated by tumor progression ($n_{\text{regressed}} = 28$, $n_{\text{progressed}} = 9$)

In terms of comorbidities, in the melanoma group, 12 patients (21.8%) had hypertension (HTN), 9 patients (16.4%) had diabetes mellitus, 8 patients (14.5%) had ischemic heart disease, and 8 patients had a history of cataract surgery. In contrast, within the VPT/RCH group, only 2 patients (8%) had HTN, and 1 patient had a history of cataract surgery, with no significant relationship observed between any demographic variables and the outcome of brachytherapy in either group [Table 1].

Among the melanoma tumors, the most common locations were the nasal region (11 tumors, 20.0%), temporal region (10 tumors, 18.2%), superior nasal region (9 tumors, 16.4%), and

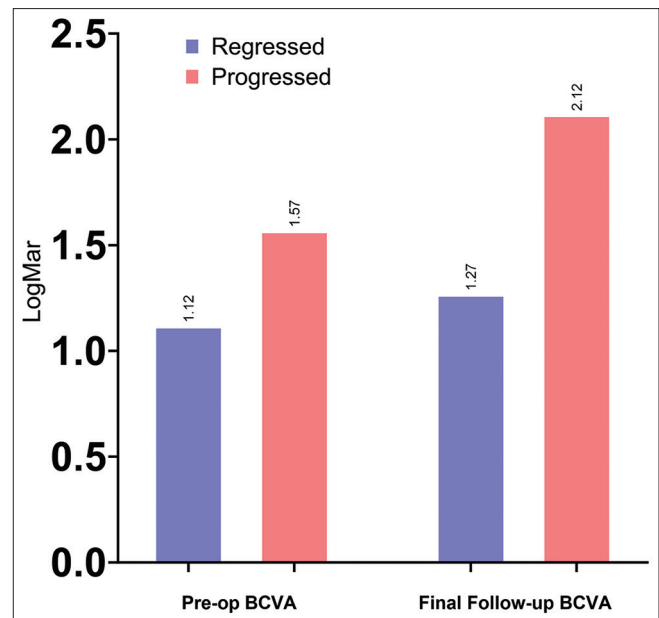


Figure 3: Mean pre- and final follow-up best-corrected visual acuity (logMAR) in the melanoma group, separated by tumor progression ($n_{\text{regressed}} = 28$, $n_{\text{progressed}} = 9$)

inferior region (8 tumors, 14.5%). For VPT/RCH tumors, the highest number was located in the inferior temporal region (13 tumors, 52.0%) and inferior region (8 tumors, 32.0%). In melanoma, there was a significant relationship between the location of the tumor and the success of brachytherapy ($P = 0.023$), with the highest rate of progress observed in tumors located in the superior temporal region (3 out of 4 patients), superior nasal region (4 out of 9 patients), and temporal region (3 out of 10 patients) [Table 1 and Figure 1b]. In the melanoma group, while 89.7% (26 out of 31) of the tumors measuring <6.4 mm were regressed, only 57.7% (15 out of 26) of those larger than 6.4 mm showed regression. There was a significant correlation between tumor thickness and brachytherapy outcomes in melanoma cases ($P = 0.007$). With respect to tumor length, although 89.5% (17 out of 19) of tumors measuring <9.75 mm showed regression, only 65.5% (19 out of 29) of those exceeding this length exhibited a similar response. Although progressed tumors tended to be larger, this relationship was not statistically significant ($P = 0.061$) [Table 1 and Figure 2a].

In the melanoma group, the most common types of plaques used were CCB (16 plaques, 29.1%) and CGD (14 plaques, 25.5%). In the VPT/RCH group, CCA plaques were used most frequently (13 plaques, 54.2%), followed by CCB plaques (10 plaques, 41.7%). The prescribed dose (Gy) was 85.83 ± 24.09 in the melanoma group and 97.08 ± 5.61 in the VPT/RCH group, with no significant relationship observed between the prescribed dose and progression of tumor in the melanoma group. The duration of plaque brachytherapy was 140.80 ± 83.71 h in the melanoma group and 84.88 ± 39.84 h in the VPT/RCH group. Regarding other variables related to brachytherapy, the dose rate at the apex was 1.05 ± 1.24 Gy/h

Table 1: Demographic and clinical characteristics of the patients undergoing brachytherapy

Variables	Melanoma (n=55), frequency (%)	P*	VPT + RCH (n=25), frequency (%)
Demographic and clinical			
Age	57.87±16.19	0.728	35.84±10.63
Sex (male)	30 (54.5)	0.297	15 (60.0)
HTN (yes)	12 (21.8)	0.119	2 (8.0)
DM (yes)	9 (16.4)	0.055	0
IHD (yes)	8 (14.5)	0.078	0
History of cataract surgery	8 (14.5)	0.337	1 (4.0)
Eye side (OD)	39 (69.1)	0.797	10 (40.0)
Tumor location			
Conjunctiva	1 (1.8)	0.023	0
Superior	2 (3.6)		1 (4.0)
Inferior	8 (14.5)		8 (32.0)
Nasal	11 (20.0)		0
Temporal	10 (18.2)		3 (12.0)
Superior nasal	9 (16.4)		1 (4.0)
Superior temporal	4 (7.3)		1 (4.0)
Inferior nasal	6 (7.3)		1 (4.0)
Inferior temporal	5 (9.1)		13 (52.0)
Tumor thickness (≥6.40 mm)	26 (47.3)	0.007	1 (4.0)
Tumor diameter (≥9.75 mm)	29 (52.7)	0.061	7 (28.0)
Brachytherapy-related			
Plaque type			
CCB	16 (29.1)	0.257	10 (41.7)
CCA	7 (12.7)		13 (54.2)
CGD	14 (25.5)		1 (4.2)
CIA	11 (20.0)		0
COB	6 (10.9)		0
Prescribed dose (Gy), mean±SD	85.83±24.09	0.427	97.08±5.61
Total time implantation (h), mean±SD	140.80±83.71	0.035	84.88±39.84
Decay factor, mean±SD	0.72±0.15	0.937	0.67±0.14
Plaque activity (MBq), mean±SD	12.44±5.66	0.828	10.42±4.57
Dose rate of the apex (Gy/h), mean±SD	1.05±1.24	0.025	1.61±1.40
Dose rate of the sclera (Gy/h), mean±SD	7.17±4.74	0.206	6.29±1.47
Total apex dose (Gy), mean±SD	80.52±29.63	0.057	96.21±7.09

*Correlation with final outcome (progressed vs. regressed), Chi-square for categorical variables, and Mann–Whitney test for continuous variables.

OD: Right eye, VPT: Vasoproliferative tumor, RCH: Retinal capillary hemangioma, Gy: Gray, SD: Standard deviation, HTN: Hypertension, DM: Diabetes mellitus, IHD: Ischemic heart disease

in the melanoma group and 1.61 ± 1.40 Gy/h in the VPT/RCH group, with a significant relationship observed between the dose rate at the apex and the tumor progression in the melanoma group ($P = 0.025$) [Table 1 and Figure 2b, c].

In the melanoma group, 74.5% of tumors were regressed after brachytherapy, compared to 100% in the VPT/RCH group. The duration of follow-up was 22.59 ± 24.66 (Median: 14.0, interquartile range [IQR]: 25) months for the melanoma group and 19.08 ± 18.57 (Median: 12.5, IQR: 13.75) months for the VPT/RCH group, with no significant relationship to the tumor progression. In both groups, BCVA and IOP did not change significantly from before brachytherapy to the last follow-up. However, in the melanoma group, there was a significant relationship between preoperative BCVA and postoperative BCVA with the tumor progression ($P_{\text{preoperative}} = 0.050$, $P_{\text{postoperative}} = 0.012$) [Table 2 and Figure 3].

The rate of recurrence and the need for retreatment were 9 patients (16.4%) in the melanoma group and 2 patients (8.0%) in the VPT/RCH group. The duration of recurrence was 12.8 ± 11.53 months in the melanoma group and 12.0 ± 9.9 months in the VPT/RCH group. In the melanoma group, 11 patients required cataract surgery, with an average time from the brachytherapy to the cataract surgery being 20.18 ± 22.31 months. In the VPT/RCH group, 6 patients required cataract surgery, with an average duration of 16.17 ± 22.11 months. Among the complications, radiation retinopathy was observed in 7 cases (12.7%) in the melanoma group, followed by retinal detachment, retinal atrophy, optic atrophy, glaucoma, and scleral thinning, each with 3 cases (5.4%). In the VPT/RCH group, radiation retinopathy was the most common complication with 4 cases (16.0%), followed by retinal detachment with 2 cases (8.0%). Among patients with longer follow-up durations (≥ 24 months), radiation

Table 2: Outcomes and complications of brachytherapy in patients with intraocular tumors

Variables	Melanoma (n=55), frequency (%)	P*	VPT + RCH (n=25), frequency (%)
Outcomes			
Regressed	41 (74.5)		25 (100)
Progressed	14 (25.5)	-	0
Follow-up duration, mean±SD	22.59±24.66	0.549	19.08±18.57
Number of IVB, mean±SD	1.17±2.13	0.151	3.36±6.14
Preoperative BCVA, mean±SD	1.23±0.98	0.050	1.18±0.81
Postoperative BCVA, mean±SD	1.48±1.05	0.012	1.27±0.82
p**	0.051		0.570
Preoperative IOP, mean±SD	14.44±4.46	0.579	13.88±3.93
Postoperative IOP, mean±SD	14.37±6.82	0.716	13.29±3.01
p**	0.763		0.421
Complications			
Recurrence (yes)	9 (16.4)		2 (8.0)
Time to recurrence (months), mean±SD	12.8±11.53		12.0±9.90
Metastasis (yes)	3 (5.4)		0
Cataract surgery (yes)	11 (20.0)		6 (24.0)
Time to surgery (months), mean±SD	20.18±22.31		16.17±22.11
Radiation retinopathy	7 (12.7)		4 (16.0)
Retinal detachment	3 (5.4)		2 (8.0)
Retinal atrophy	3 (5.4)		1 (4.0)
Optic atrophy	3 (5.4)		0
Glaucoma	3 (5.4)		1 (4.0)
Scleral thinning	3 (5.4)		0
Optic neuropathy	2 (3.64)		1 (4.0)
Radiation maculopathy	1 (1.8)		0
Central macular edema	1 (1.8)		1 (4.0)

*Correlation with final outcome (progressed vs. regressed), Chi-square for categorical variables, and Mann–Whitney test for continuous variables, **Pre versus post, Wilcoxon test. VPT: Vasoproliferative tumor, RCH: Retinal capillary hemangioma, BCVA: Best-corrected visual acuity, IOP: Intraocular pressure, SD: Standard deviation, IVB: Intravitreal bevacizumab

retinopathy was observed in 5 out of 24 patients (20.8%), including cases from both the melanoma and VPT/RCH groups [Table 2].

DISCUSSION

The primary finding of this study is the significant efficacy of Ru-106 plaque brachytherapy in treating various types of intraocular tumors, including melanoma, VPTs, and RCH. Specifically, the study demonstrates a substantial tumor regression rate postbrachytherapy, with 74.5% of melanoma tumors and 100% of VPT/RCH tumors regressing. The treatment's success was significantly influenced by factors such as tumor diameter, thickness, and dose rate at the apex in the melanoma group, with lower thickness and higher dose rates in melanoma tumors being associated with better outcomes. Furthermore, the study highlights the relationship between the tumor's location and brachytherapy success in melanoma, particularly noting higher progression rates in the superior temporal, superior nasal, and temporal regions.

The findings of the study regarding melanoma are generally consistent with the existing literature on brachytherapy's effectiveness. Cennamo *et al.* (2022) evaluated Ru-106 plaque brachytherapy for uveal melanoma in 350 patients, reporting

excellent tumor regression with an average thickness reduction from 4.52 ± 1.78 mm to 1.75 ± 0.21 mm post-treatment and local control rates exceeding 95% over a follow-up of up to 9 years.¹⁷ Similarly, Takiar *et al.* (2014) found a high local control rate of 97% at 5 years using Ru-106 plaques for uveal melanoma, emphasizing the efficacy of Ru-106 brachytherapy in achieving significant tumor regression and minimal toxicity.¹⁸

However, the regression rate observed in our melanoma cohort (74.5%) was somewhat lower than rates reported in these studies. This discrepancy might be attributed primarily to differences in tumor characteristics. Notably, in the studies by Cennamo *et al.* and Takiar *et al.*, tumors treated had lower average thicknesses (approximately 4.5 mm and 3.05 mm, respectively), whereas the average tumor thickness in our study was higher (6.38 mm). Moreover, Pilotto *et al.* (2009) reported tumor regression rates of 98% with iodine-125 brachytherapy in small choroidal melanomas (mean thickness of approximately 2.2 mm),¹¹ underscoring how smaller tumors typically demonstrate more robust regression following treatment. Thus, the larger baseline tumor size in our study likely contributed significantly to the comparatively lower regression rates observed.

In the VPT/RCH group, the high regression rate (100%) after brachytherapy is consistent with the successful outcomes reported in other studies. Naseripour *et al.* evaluated the effect of Ru-106 plaques in controlling retinal VPTs and found significant anatomical and functional improvements postbrachytherapy. Their study reported a complete reabsorption of significant exudative retinal detachments in the majority of patients and highlighted the effectiveness of high-dose Ru-106 plaques in treating these tumors.¹⁹ Adding to this, the study by Abolfathzadeh *et al.*²⁰ further supports the effectiveness of Ru-106 plaque brachytherapy in treating VPTs. In their 15-year single-center experience involving 25 patients, the mean tumor thickness decreased significantly from 4.27 ± 1.10 mm to 2.60 ± 0.63 mm postoperatively ($P < 0.001$), with complete resolution of subretinal fluid in 81.80% of cases. The study also reported that visual acuity of more than 20/400 was maintained in 60% of patients at the last follow-up, indicating that Ru-106 brachytherapy is a safe and effective treatment method for VPTs. This similarity underscores brachytherapy's effectiveness in managing various ocular tumors, particularly those with a vascular component.

The current study identified a significant relationship between tumor location and brachytherapy success in the melanoma group. The melanoma tumors located in the superior temporal, superior nasal, and temporal regions showed a higher rate of tumor progression. This finding is corroborated by Fili *et al.*, who found that the proximity of choroidal melanomas to the optic disc influenced treatment outcomes, with specific locations posing higher risks for treatment failure.²¹ The varying rates of tumor regression in different anatomical locations can be attributed to several factors. These include the anatomical complexity of the area, challenges in accessing the tumor, and difficulties in placing the brachytherapy plaque accurately. Ensuring the correct placement of the plaque is crucial for delivering the appropriate radiation dose to the tumor.

In the present study, within the melanoma group, tumor thickness and diameter were higher in progressed tumors compared to regressed tumors. This difference was statistically significant for tumor thickness. In a study conducted by Messer *et al.*, larger tumor size was identified as a negative prognostic factor. The reason is that the delivery and penetration of therapeutic agents might be less effective in larger tumors, making complete eradication more challenging.²²

The dose rate at the apex showed a significant relationship with tumor progression in the melanoma group. This finding is consistent with the detailed dosimetric analyses performed by Fili *et al.*, which emphasized the importance of precise dose delivery to achieve optimal outcomes.²¹ Ensuring adequate dose rates is crucial for maximizing brachytherapy efficacy while minimizing adverse effects.

In our study, radiation retinopathy and maculopathy were identified in 12.7% and 1.8% of melanoma patients, respectively. These rates are much lower than those seen

in many larger studies with extended follow-up periods. Gündüz *et al.* reported a 42% incidence of nonproliferative radiation retinopathy at 5 years postplaque radiotherapy for posterior uveal melanoma,²³ whereas Shields *et al.* identified retinopathy in 39% and maculopathy in 18% of patients at 5 years following combined I-125 plaque and TTT.¹⁶ Finger *et al.* reported a cumulative 5-year incidence of 56.8% for radiation maculopathy in patients treated with Pd-103 plaques.²⁴ Bianciotto *et al.* also documented a 5.8% 5-year rate of proliferative radiation retinopathy, which rose to 7% at 10 and 15 years.²⁵ The lower rates in our study may be explained by the shorter follow-up duration, as radiation retinopathy is a progressive complication that increases with time. Moreover, variations in diagnostic methodology likely affected detection. The selection of radioisotopes is a critical consideration, Ru-106 in our analysis compared to I-125, Pd-103, or Ir-192 in other studies, since each exhibits unique tissue penetration properties and dose distributions. Differences may also arise from variations in tumor size, thickness, and proximity to critical structures. Alongside radiation retinopathy, our study identified other complications, including retinal detachment and cataract formation, which align with existing literature findings. Studies conducted by Cuffaro *et al.*,²⁶ Takiar *et al.*,¹⁸ and Peddada *et al.*⁶ similarly identified adverse effects. Hasegawa *et al.* reported that radiation retinopathy was the most prevalent complication following brachytherapy for uveal melanoma (54.7%), with optic neuropathy (20.4%) and vitreous hemorrhage (6.7%) following in frequency.²⁷ Identifying these potential complications is essential for enhancing long-term visual outcomes and informing post-treatment monitoring strategies.

To mitigate radiation-associated complications, integrating adjunctive therapies, such as antivascular endothelial growth factor (anti-VEGF) agents, may be beneficial. Previous studies have demonstrated that intravitreal injections of anti-VEGF can effectively manage radiation retinopathy, reducing macular edema and neovascular complications. Prophylactic or early intervention strategies with anti-VEGF therapies should be considered in clinical practice and future research protocols to enhance patient outcomes following brachytherapy.²⁸

The study noted no significant changes in BCVA and IOP from before brachytherapy to the last follow-up. However, preoperative and postoperative BCVA showed significant relationships with tumor progression in the melanoma group. This observation is supported by long-term follow-up studies, such as those by Cennamo *et al.* (2022)¹⁷ and Jiang *et al.*,¹⁰ which also reported stable visual acuity in the majority of patients postbrachytherapy. In another study conducted by O'Day *et al.* on patients with choroidal melanoma, most patients retained vision of 6/60 or better after brachytherapy.²⁹ Conversely, in the study by Hasegawa *et al.*, patients' vision worsened at long-term follow-up after brachytherapy, and this deterioration was related to the radiotherapy dose delivered to the whole eye.²⁷ While globe salvage is the primary goal in ocular oncology, maintaining visual acuity is an important

secondary objective, as it significantly affects patients' quality of life.

The recurrence rate identified in our study for melanoma (16.4%) is consistent with previously reported rates, including the 18% documented by Quivey *et al.* for I-125 brachytherapy at 5 years.³⁰ The recurrence rate in our study is marginally elevated compared to the 10.3% reported in the COMS trial at 5 years, which may be attributed to variations in tumor characteristics and the reduced follow-up duration in our study.³¹ Shields *et al.* reported a 3% recurrence rate, mainly attributed to the use of TTT, suggesting that combined treatment modalities might enhance local control compared to Ru-106 monotherapy.¹⁶ In contrast, Fili *et al.* reported a significantly higher recurrence rate of 43% in juxtapapillary melanomas, highlighting the challenge of treating tumors with close proximity to critical anatomical structures, which requires precise plaque placement and customized radiation delivery.²¹ The recurrence rate in the VPT/RCH group (8.0%) was significantly lower, indicating the superior efficacy of Ru-106 brachytherapy in less aggressive vascular tumors relative to melanoma and other difficult sites. The comparative findings highlight the significant impact of tumor characteristics, anatomical location, radiation modality, and adjunctive treatments on recurrence outcomes in brachytherapy for intraocular tumors.

A notable limitation of this study is the considerable heterogeneity within the patient cohort. This diversity included various tumor types, sizes, locations, and biological behaviors. Such heterogeneity might introduce variability in treatment responses, complicating the generalization of results. Future prospective studies should aim to categorize patients more homogeneously or use stratified analysis methods to better understand tumor-specific responses to brachytherapy. Moreover, considering variables like tumor morphology can be helpful.

In addition, the single-center and retrospective design of this study may introduce selection and information bias, potentially limiting the external validity of our findings. Prospective multicenter studies with larger, more diverse populations are needed to confirm these findings and enhance generalizability.

This study provides valuable insights into the outcomes of brachytherapy for various ocular tumors, emphasizing the treatment efficacy and associated complications. The high rates of tumor regression, particularly in the VPT/RCH group, underscore potential of brachytherapy as a primary treatment modality. The significant relationships between tumor characteristics, such as location, thickness, diameter, and dose rate at the apex, and treatment outcomes highlight the need for tailored approaches in brachytherapy planning. Comparing these findings with existing literature affirms the study conclusions and underscores the critical factors influencing brachytherapy success. Future research should continue to refine brachytherapy techniques and explore strategies to mitigate complications, ultimately improving patient outcomes in ocular oncology.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

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